

001 · 1 毫秒的问题 · The 1 ms problem 0:00

两个机器人关节，在同一时刻收到同样的命令。

Two robot joints get the same command at the same time.

它们真的会在同一瞬间动起来吗？

Will they really move at the same moment?



002 · 1 毫秒的问题 · The 1 ms problem 0:11

这门课讲 EtherCAT，很多机器人关节里用的就是这条总线。

This course is about EtherCAT, the bus inside many robot joints.

讲它怎么工作之前，先回答一个问题：为什么时间这么要紧?

Before how it works, one question: why does timing matter so much?



003 · 我们的总线 · Our line 0:26

每一集都用同一条小总线。

Every episode uses the same small line.

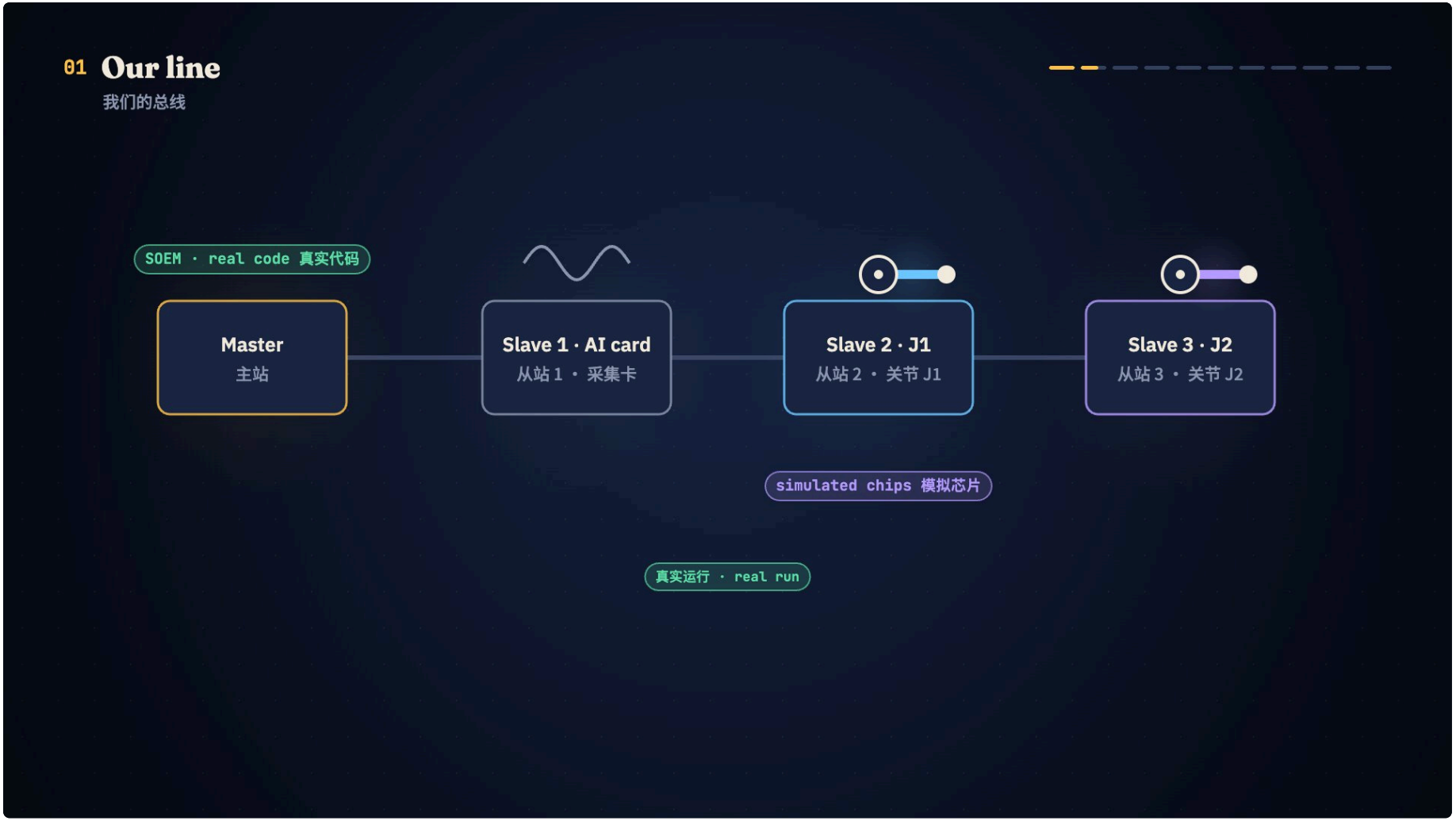
一个主站，后面一根网线串着三个从站。

One master, then three slaves on one cable.

主站：发命令的控制器。从站：听命令、回数据的设备。

Master: the controller that sends commands. Slave: a device that obeys and reports.

从站 1 是模拟量采集卡，从站 2 和 3 是两个关节驱动器：J1、J2。



004 · 我们的总线 · Our line 0:51

主站是真实的开源代码 SOEM，从站是模拟出来的芯片。

The master is real open-source code, SOEM. The slaves are simulated chips.

这门课里的每个数字，都来自真实运行。

Every number in this course comes from running it.



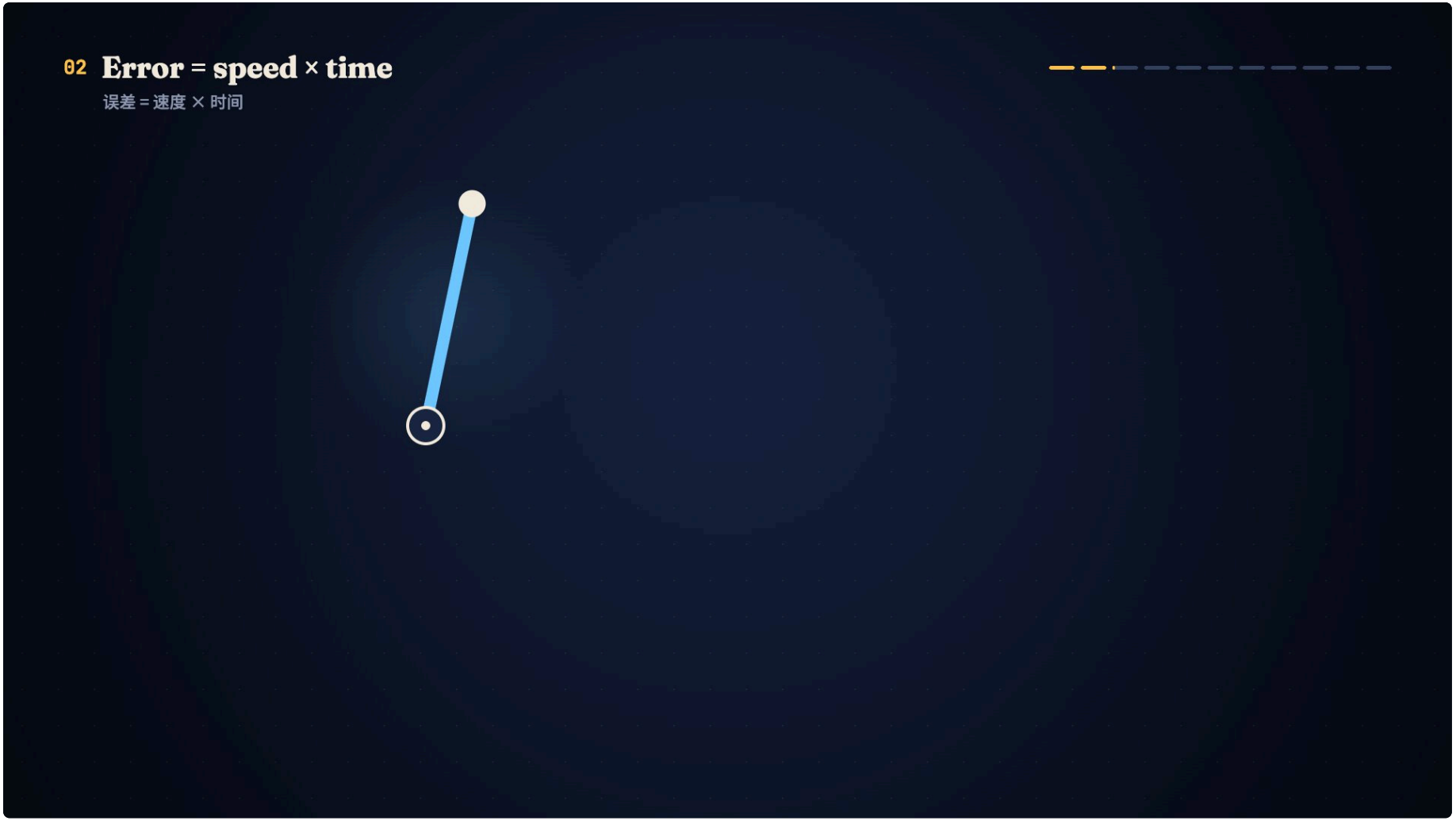
005 · 我们的总线 · Our line 1:02

主站每 1 毫秒给 J1、J2 发一个新的目标位置。

Every 1 ms, the master sends J1 and J2 a new target position.

两个关节走同一条缓慢的正弦曲线，所以它们应该永远在同一个位置。

Both follow the same slow wave, so they should always be in the same place.



006 · 误差 = 速度 × 时间 · Error = speed × time 1:19

运动中的关节不会等你。

A moving joint does not wait for you.



007 · 误差 = 速度 × 时间 · Error = speed × time 1:24

目标晚一点生效，关节就落后：落后量 = 速度 × 晚了多久。

Apply a target late, and the joint is behind by speed × lateness.



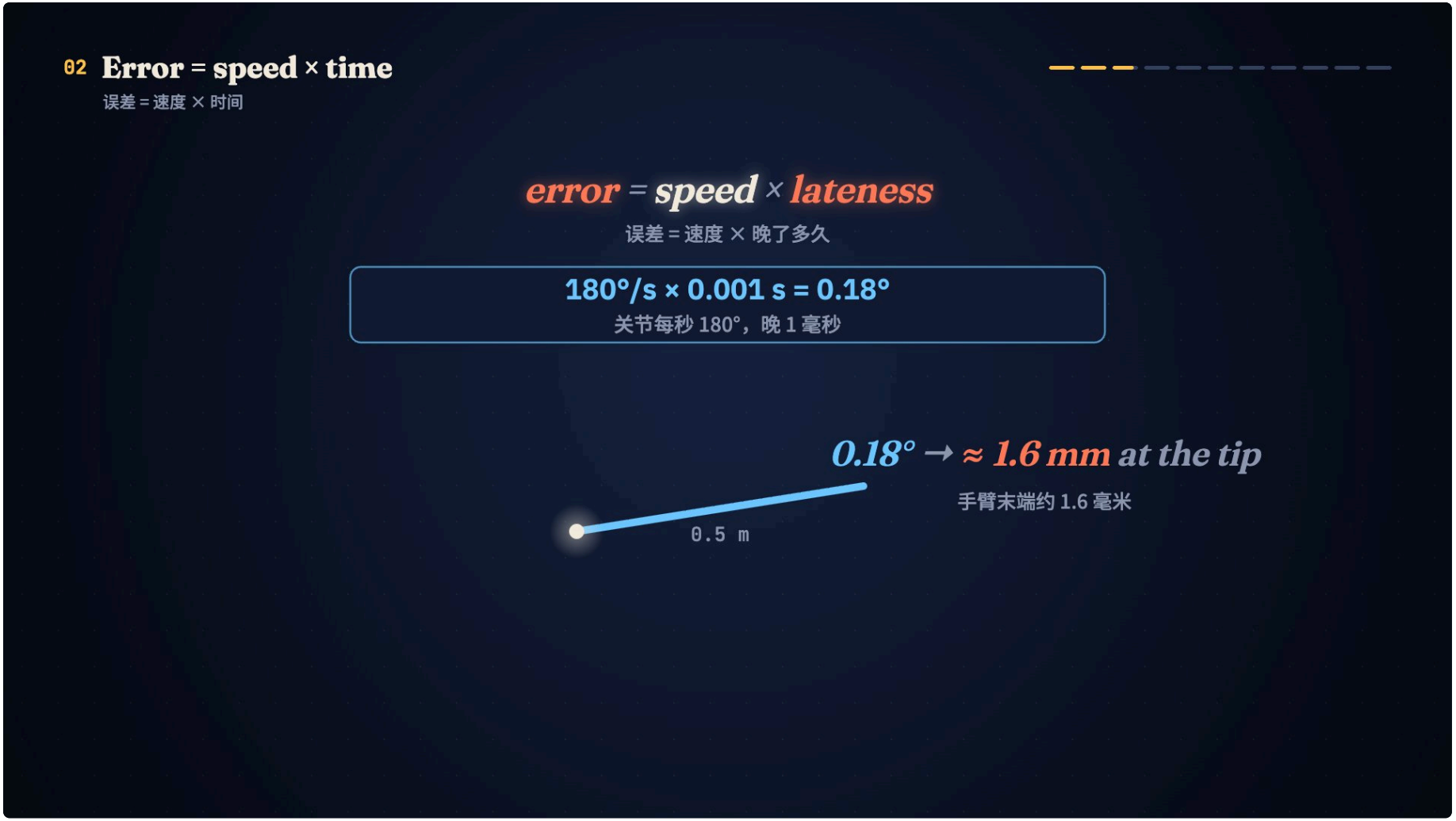
008 · 误差 = 速度 × 时间 · Error = speed × time 1:31

机床走 F1000：每分钟 1000 毫米，约每秒 16.7 毫米。

A machine tool at F1000 moves 1000 mm per minute: about 16.7 mm per second.

晚 1 毫秒：16.7 mm/s × 0.001 s ≈ 0.017 mm，约 17 微米。

1 ms late: 16.7 mm/s × 0.001 s ≈ 0.017 mm, about 17 μm.



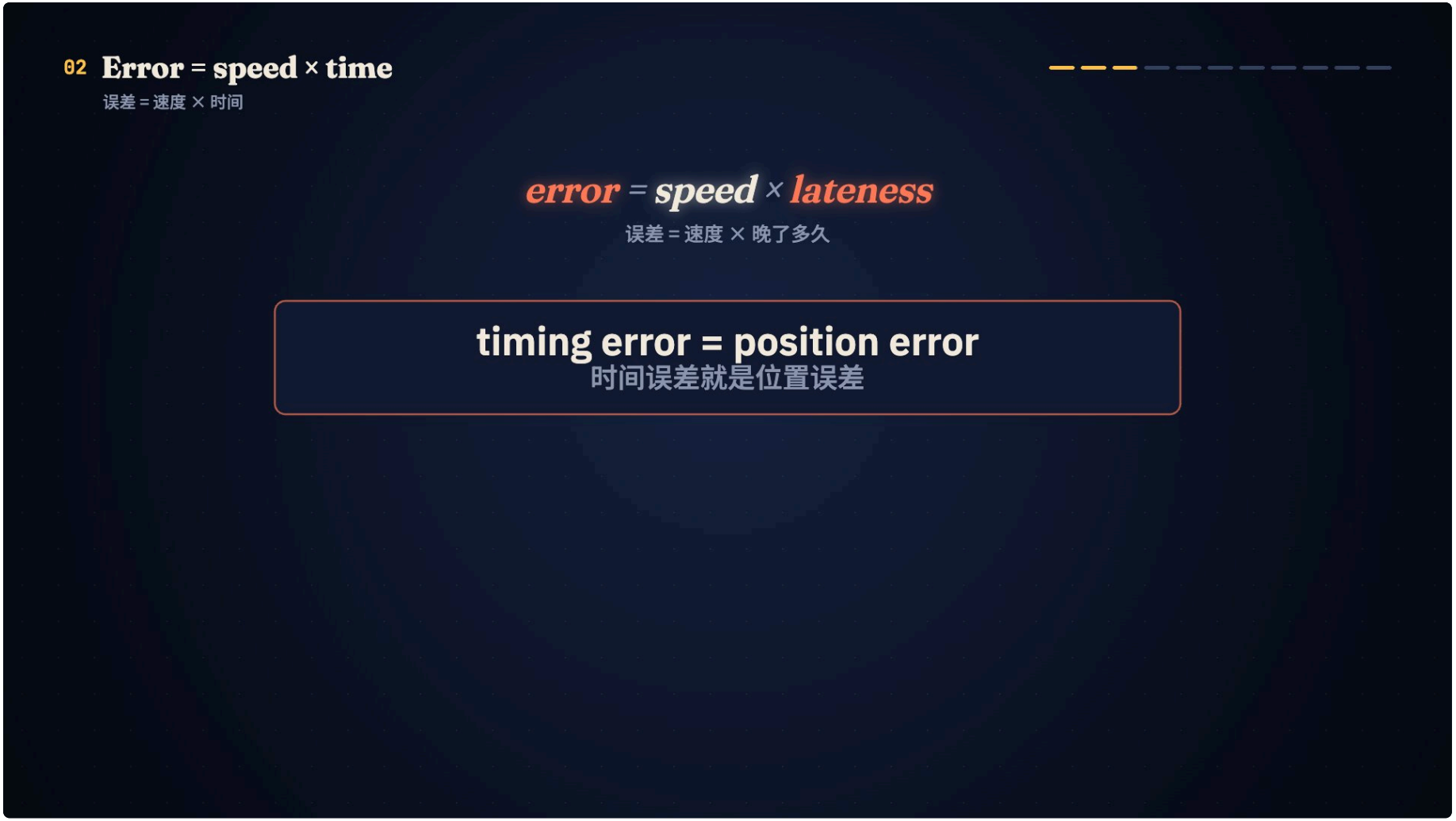
009 · 误差 = 速度 × 时间 · Error = speed × time 1:46

机器人关节每秒转 180°：晚 1 毫秒就是 0.18°。

A robot joint turning 180° per second: 1 ms late is 0.18°.

在 0.5 米长的手臂末端，0.18° 约等于 1.6 毫米。

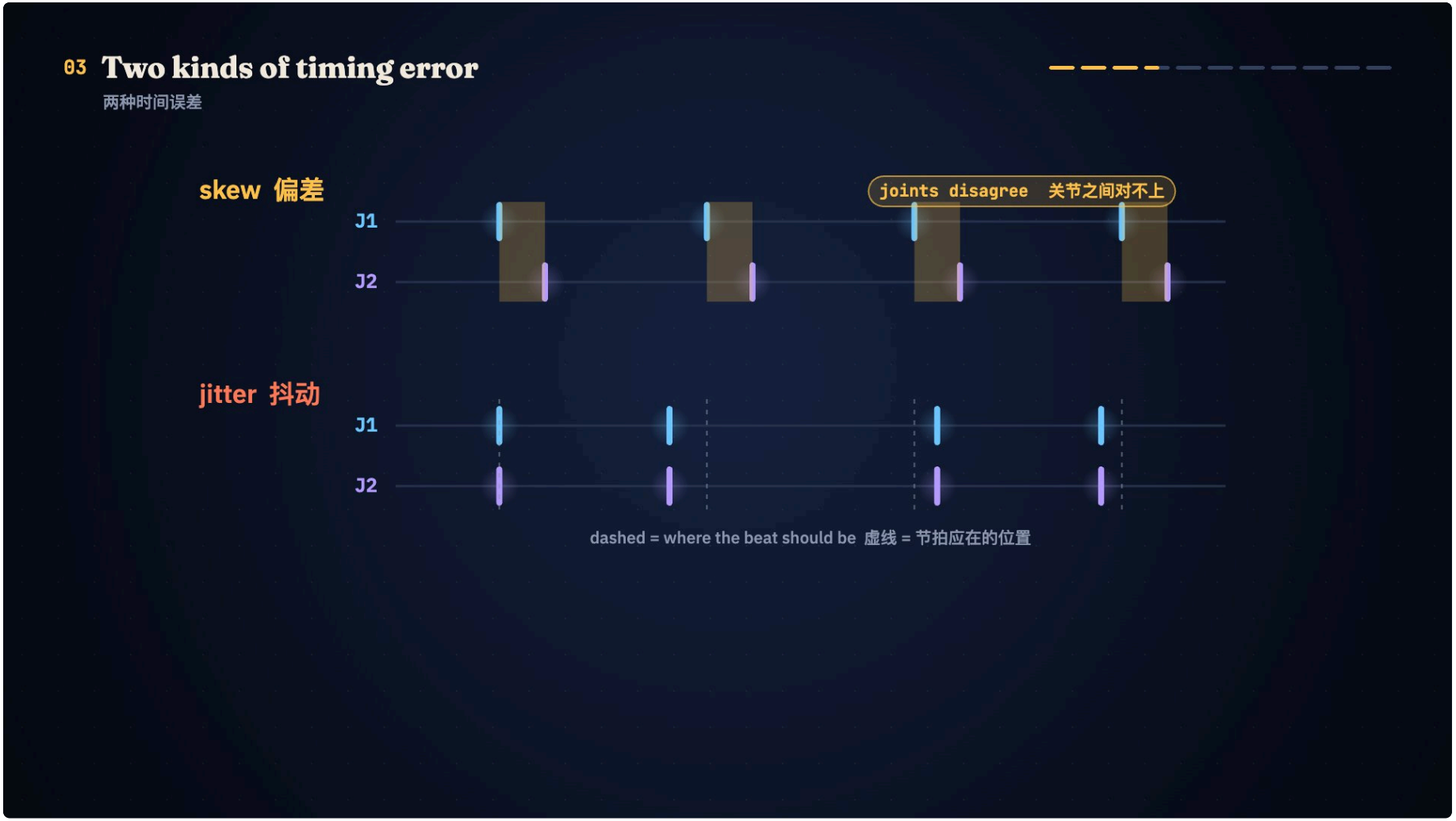
At the end of a 0.5 m arm, 0.18° is about 1.6 mm.



010 · 误差 = 速度 × 时间 · Error = speed × time 1:59

对运动中的机器来说，时间误差就是位置误差。

For a moving machine, a timing error is a position error.



011 · 两种时间误差 · Two kinds of timing error 2:07

时间可以错在两个地方。

Timing can go wrong in two different ways.

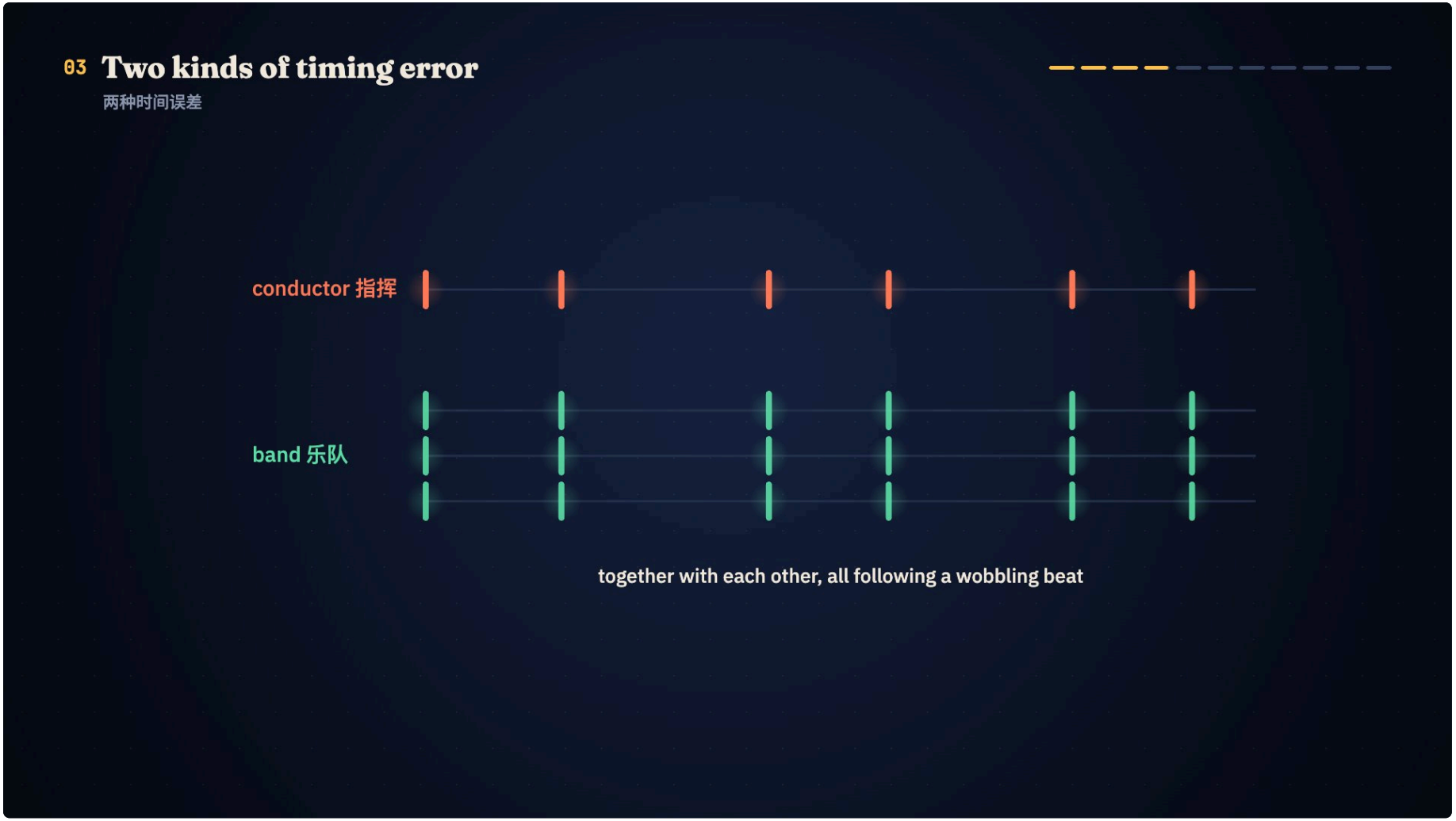
偏差（skew）：同一个周期里，J1 和 J2 动作的时刻不一样。

Skew: within one cycle, J1 and J2 act at different moments.

抖动（jitter）：周期本身在晃。1 毫秒一会儿变 0.8，一会儿变 1.2。

Jitter: the cycle itself wobbles. 1 ms becomes 0.8, then 1.2.

偏差让关节之间对不上



012 · 两种时间误差 · Two kinds of timing error 2:28

抖动让整个动作磕磕绊绊，哪怕关节彼此一致。

Jitter makes the whole motion stumble, even when the joints agree.

像一支乐队：大家彼此很齐，但指挥的拍子忽快忽慢。

Like a band that plays together, but follows a conductor whose beat wobbles.



013 · CANopen 的做法 · How CANopen handles it 2:43

如果挨个给电机发命令，第一个已经动了，最后一个才收到。

Send each motor its command in turn, and the first moves before the last one hears.

这就是偏差。CANopen 的 CSP 模式用一条广播来解决。

That is skew. CANopen's CSP mode fixes it with one broadcast.

先把目标分别发给每个驱动器，再发一条 SYNC 报文（ID 0x80）：大家一起动。

First each drive gets its target. Then one SYNC message, ID 0x80, says: everyone, go.

多报文同时到达所有驱动器，偏差就小了



014 · CANopen 的做法 · How CANopen handles it 3:25

每个驱动器还要经过中断才响应，又添一段延迟。

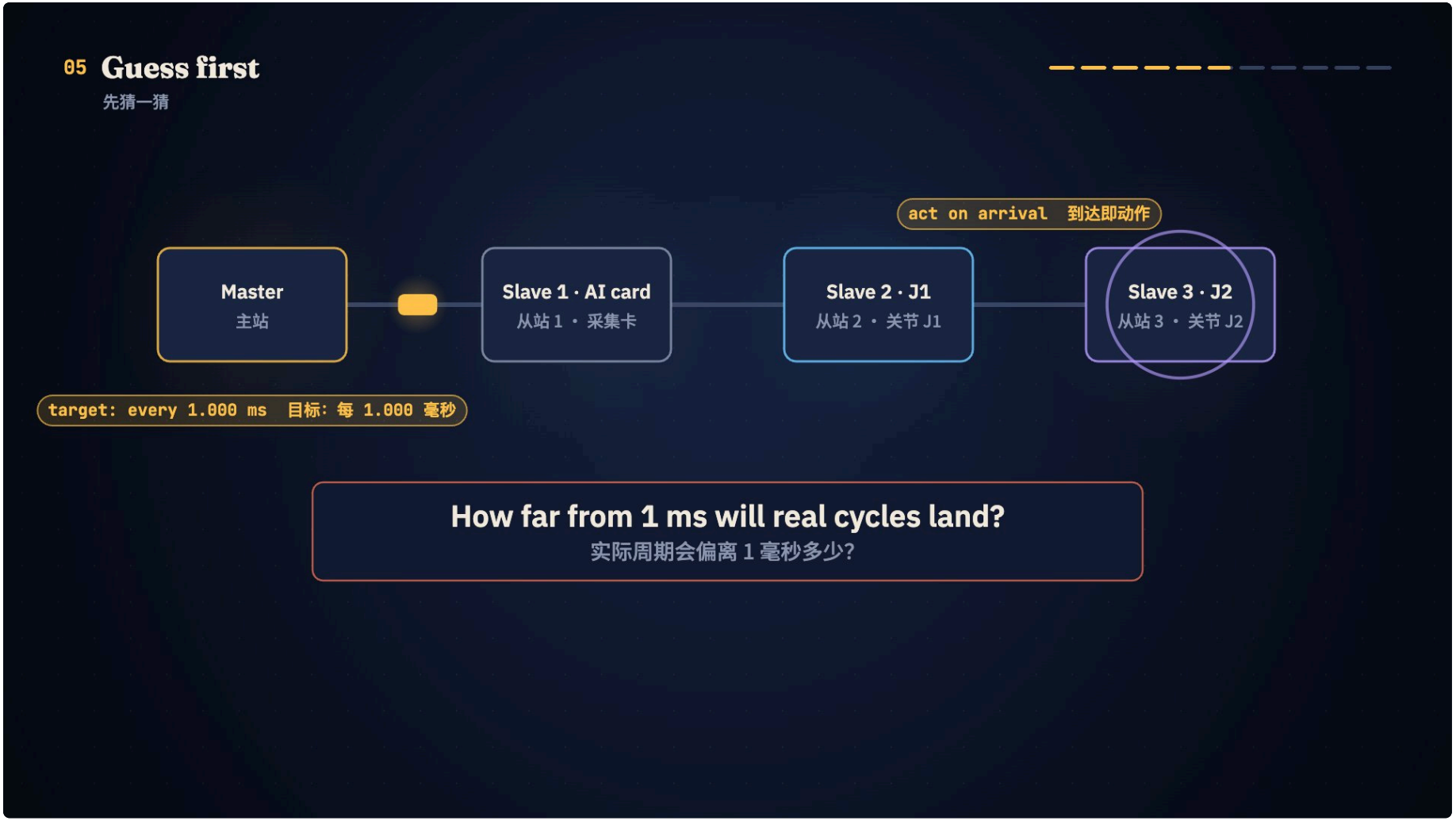
Then each drive reacts through an interrupt, which adds its own delay.



015 · CANopen 的做法 · How CANopen handles it 3:32

所以 SYNC 治了偏差，治不了抖动：每个驱动器都继承了主站的晃动。

So SYNC fixes skew, not jitter. Every drive inherits the master's wobble.



016 · 先猜一猜 · Guess first 3:41

回到我们的总线。先让驱动器在新目标到达的那一刻动作。

Back on our line. First, the drives act the moment new targets arrive.

主站尽量每 1 毫秒准时发一次。

The master tries to send exactly every 1 ms.

猜一猜：实际的周期会偏离 1 毫秒多少?

Guess: how far from 1 ms will the real cycles land?



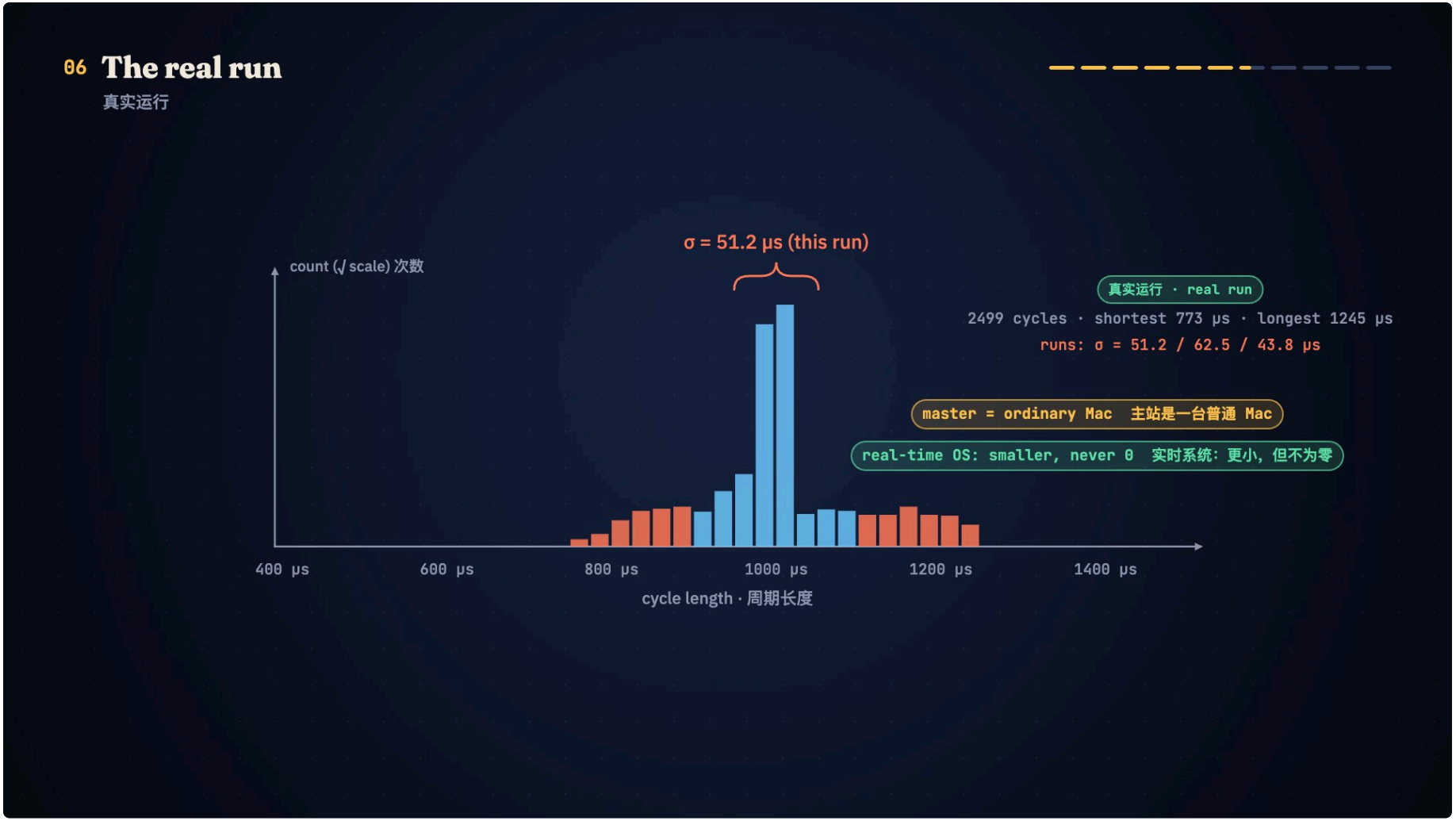
017 · 真实运行 · The real run 4:03

运行 3 秒：在 J1 上测到约 2500 个周期。

3 seconds of running: about 2,500 cycles measured at J1.

大多数周期在 1 毫秒附近，但总有一些早到或晚到几百微秒。

Most cycles stay near 1 ms. But some come hundreds of microseconds early or late.



018 · 真实运行 · The real run 4:16

离散程度（一个标准差）每次运行都是几十微秒。

The spread, one standard deviation, is tens of microseconds, in every run.

这些晃动来自主站：一台普通的 Mac，不是实时系统。

The wobble comes from the master: an ordinary Mac, not a real-time system.

换成实时操作系统能小很多，但永远不会是零。

A real-time OS makes it much smaller, but never zero.



019 · 真实运行 · The real run 4:35

再看 J1 和 J2 之间：每个周期都相差 310 纳秒。

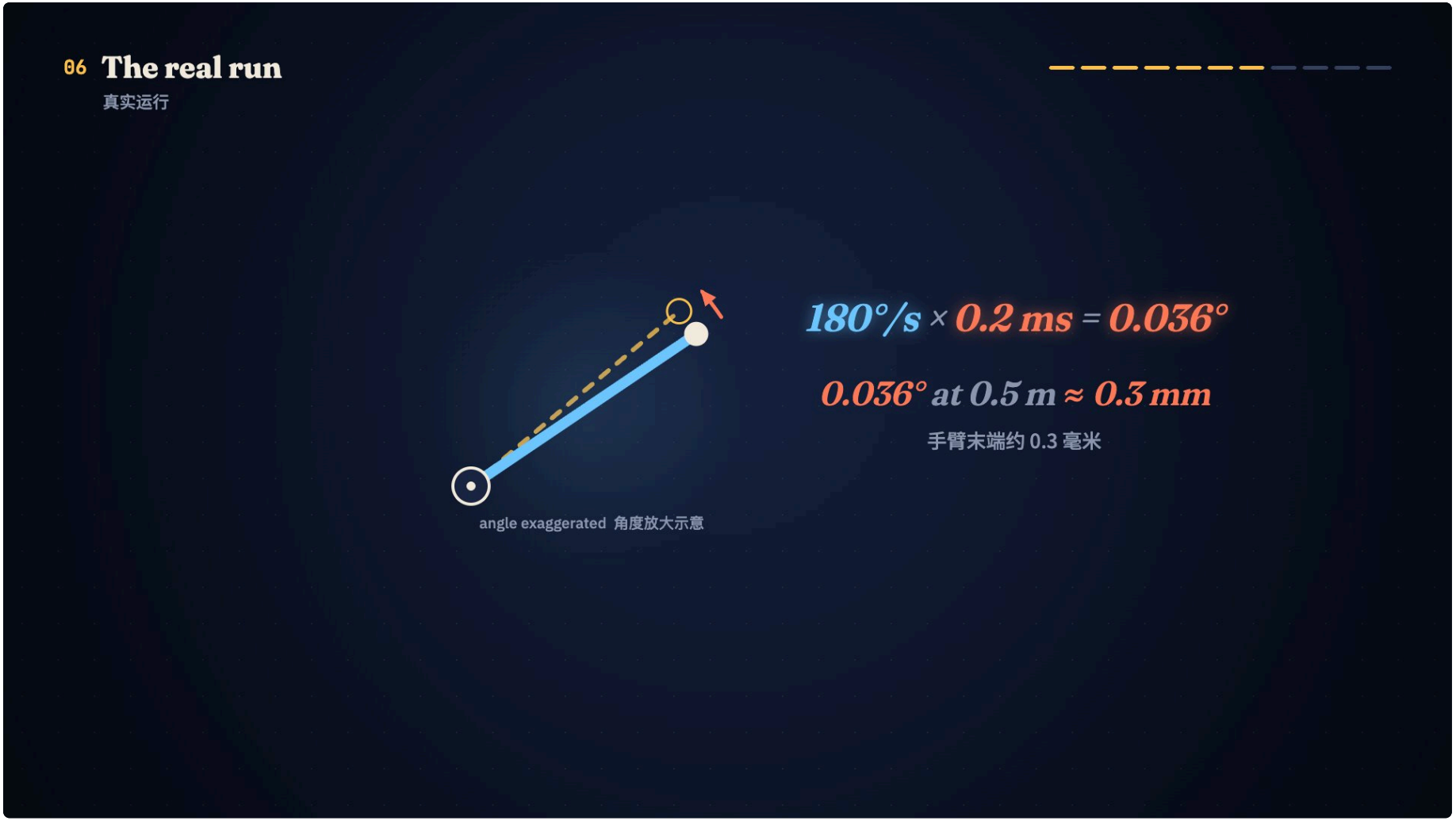
Now J1 against J2: they act 310 ns apart, in every cycle.

310 纳秒是模拟的网线加芯片延迟：同一帧先经过 J1，再经过 J2。

310 ns is the simulated cable and chip delay. One frame passes J1, then J2.

所以偏差很小。一帧怎么做到这一点，第 3 集再讲。

So skew is tiny. Episode 3 shows how one frame does that.



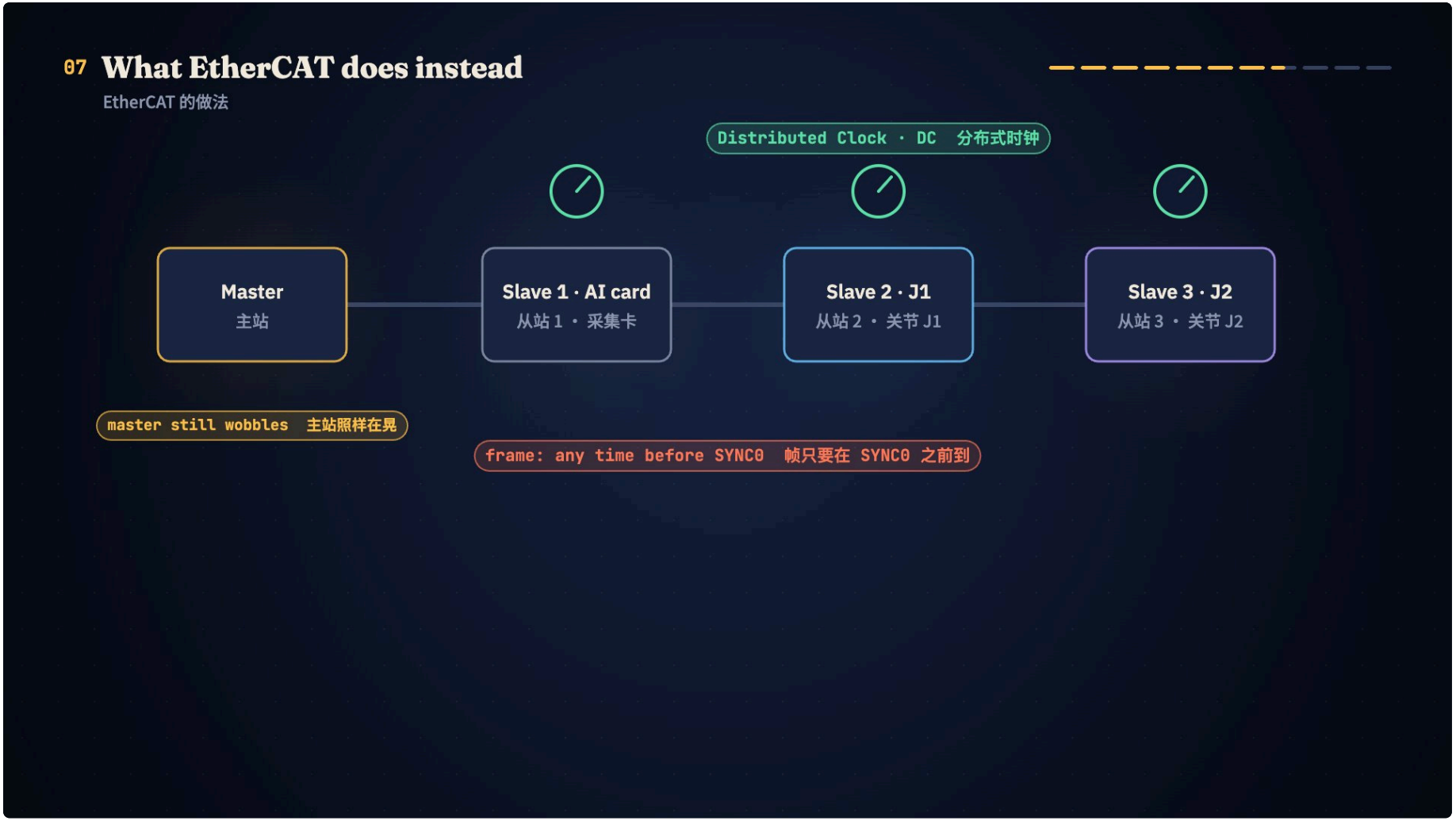
020 · 真实运行 · The real run 4:55

可是抖动原封不动地传给了电机。

But the jitter goes straight through to the motors.

以每秒 180° 计，晚到 0.2 毫秒的周期让 J1 落后 0.036°，在手臂末端约 0.3 毫米。

At 180°/s, a cycle 0.2 ms late leaves J1 0.036° behind: about 0.3 mm at the arm tip.



021 · EtherCAT 的做法 · What EtherCAT does instead 5:12

同一条总线，只改一处：每个驱动器按自己的时钟动作。

Same line, one change: each drive acts on its own clock.

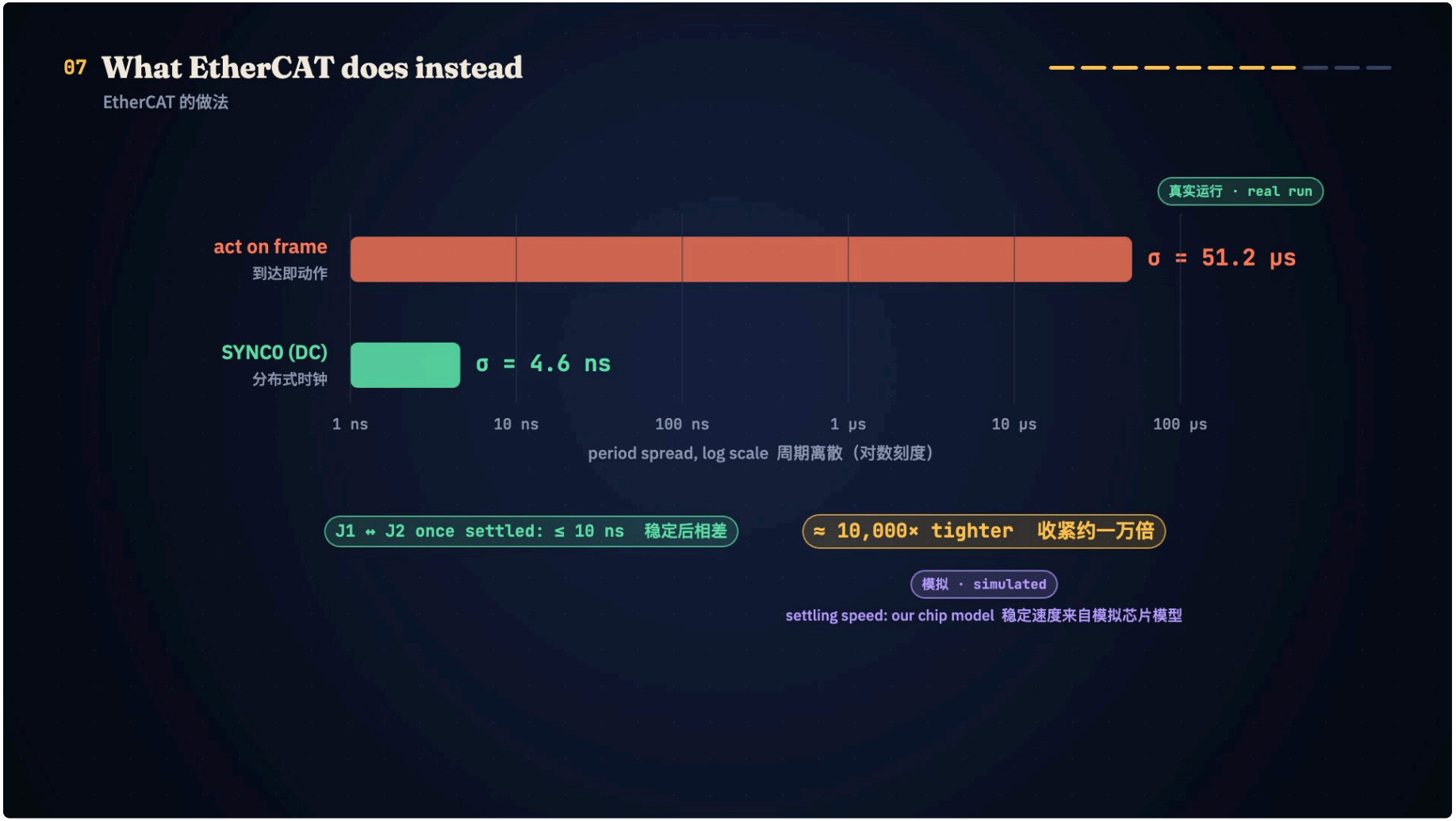
每个从站芯片里都有一个时钟。EtherCAT 让这些时钟步调一致。

Every slave chip has a clock. EtherCAT keeps all those clocks in step.

这叫分布式时钟（Distributed Clock），简称 DC。

This is called the Distributed Clock, DC for short.

每个驱动器按自己的时钟，每 1 毫秒产生一个叫 SYNC0 的脉冲，就在这一刻动作。



022 · EtherCAT 的做法 · What EtherCAT does instead 5:50

驱动器上的周期离散：约 5 纳秒，每次运行都一样。

Cycle spread at the drives: about 5 ns, in every run.

时钟稳定下来以后，J1 和 J2 动作的时刻相差在 10 纳秒左右。

Once the clocks settle, J1 and J2 act within about 10 ns of each other.

从几十微秒到几纳秒：大约收紧了一万倍。

From tens of microseconds to a few nanoseconds: about ten thousand times tighter.

注意：时钟多快能稳定，取决于我们模拟的芯片，不是某款真实芯片的数据。



023 · 这是怎么做到的? · How is that possible? 6:21

要做到这一点，需要回答四个问题，一个一个来。

Getting there takes four questions, one at a time.

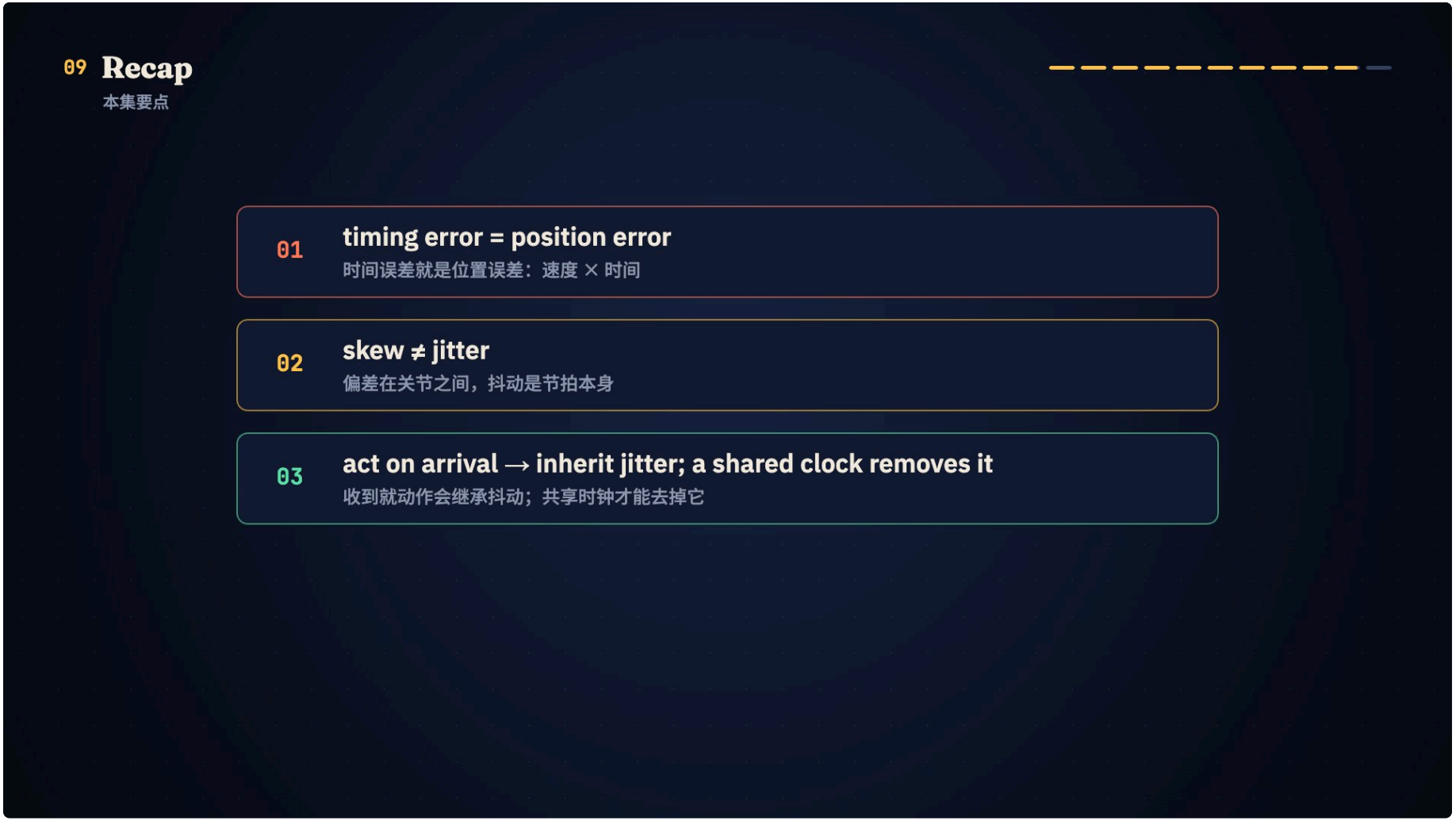
一根网线，怎么就已经是一个环？第 2 集。

How is one cable already a ring? Episode 2.

一帧怎么不停下来，就服务了每一个从站？第 3 集。

How does one frame serve every slave without stopping? Episode 3.

总线怎么启动，怎么约定谁收发哪些数据？第 4 集。



024 · 本集要点 · Recap 6:54

一、对运动中的机器，时间误差就是位置误差：速度 × 时间。

One: for a moving machine, timing error is position error. Speed × time.

二、偏差和抖动是两回事：一个在关节之间，一个是节拍本身。

Two: skew and jitter differ. One is between joints, the other is the beat itself.

三、收到报文就动作，会继承发送方的抖动；共享的时钟才能去掉它。

Three: acting when a message arrives inherits the sender's jitter. A shared clock removes it.



025 · 你来算 · Your turn 7:18

一个关节每秒转 90°，某个周期晚到了 0.3 毫秒。

A joint turns at 90° per second. One cycle arrives 0.3 ms late.

它落后了多少度?

How far behind is it, in degrees?

90°/s × 0.0003 s = 0.027°。

90°/s × 0.0003 s = 0.027°.



026 · 你来算 · Your turn 7:36

下一集：一根网线，为什么它已经是一个环。

Next: one cable, and why it is already a ring.

练习 · Practice

先自己答，再看答案。每个答案都写了依据：源码行，或者真实运行。 · Answer first, then open the answer. Each one says what it rests on: source lines or a real run.

第 1 题 · Q1 视频里的题 · From the video 3:40

驱动器在新目标到达时动作，主站尽量每 1 毫秒发一次。实际周期会偏离 1 毫秒多少？

The drives act when new targets arrive, and the master tries to send every 1 ms. How far from 1 ms do the real cycles land?

大多数在 1 毫秒附近，但标准差有几十微秒，总有周期早到或晚到几百微秒，偶尔还会晚到几毫秒（macOS 调度的停顿）。J1 和 J2 之间却每个周期都只差 310 纳秒。

Most stay near 1 ms, but the spread is tens of microseconds, some cycles come hundreds of microseconds early or late, and now and then one is milliseconds late (a macOS scheduling stall). Yet J1 and J2 are only 310 ns apart in every cycle.

依据 · Evidence

三次真实运行，每次约 2500 个周期，data.py 逐条检查这些性质。310 纳秒是模拟的网线（5 ns）加芯片处理（305 ns，取自 ET1100 数据手册）延迟。 / Three real runs of about 2,500 cycles each; data.py checks each property. The 310 ns is the simulated cable (5 ns) plus chip processing delay (305 ns, from the ET1100 datasheet).

`episodes/ep01-sync-is-accuracy/data/runs.sh`

第 2 题 · Q2 视频里的题 · From the video 按代码推演 · Worked out from the code 7:17

一个关节每秒转 90°，某个周期晚到了 0.3 毫秒。它落后了多少度?

A joint turns at 90° per second. One cycle arrives 0.3 ms late. How far behind is it, in degrees?

90°/s × 0.0003 s = 0.027°。

90°/s × 0.0003 s = 0.027°.

依据 · Evidence

误差 = 速度 × 晚了多久。先把毫秒换成秒。 / Error = speed × lateness. Convert ms to s first.

第 3 题 · Q3 新题 · New 按代码推演 · Worked out from the code

机床以 F3000 进给，主站抖动 50 微秒。位置误差大约是多少?

A machine tool feeds at F3000 and the master jitters by 50 µs. Roughly what position error is that?

F3000 = 3000 mm/min = 50 mm/s。50 mm/s × 0.00005 s = 0.0025 mm，也就是 2.5 微米。

F3000 = 3000 mm/min = 50 mm/s. 50 mm/s × 0.00005 s = 0.0025 mm, that is 2.5 µm.

依据 · Evidence

F 值的单位是毫米每分钟，先除以 60 换成每秒。 / The F value is in mm per minute: divide by 60 to get per second.

第 4 题 · Q4 新题 · New 按代码推演 · Worked out from the code

三个驱动器收到同一条广播就动作，但广播在 1.0、2.1、2.9 毫秒时发出。这是偏差还是抖动?

Three drives act on one broadcast, but the broadcasts leave at 1.0, 2.1 and 2.9 ms. Is that skew or jitter?

抖动。三个驱动器彼此一致（没有偏差），但节拍本身是 1.1 毫秒、0.8 毫秒，不是 1 毫秒。

Jitter. The three drives agree with each other (no skew), but the beat itself is 1.1 ms, then 0.8 ms, not 1 ms.

依据 · Evidence

偏差是同一周期里关节之间的时刻差；抖动是周期本身的长短在变。 / Skew is the time difference between joints within one cycle; jitter is the cycle length itself changing.

第 5 题 · Q5 新题 · New

用了分布式时钟以后，主站发送仍然晃几十微秒。为什么这些晃动传不到关节?

With distributed clocks, the master's sending still wobbles by tens of microseconds. Why doesn't that wobble reach the joints?

关节不再在帧到达时动作，而是在自己时钟产生的 SYNC0 时刻动作。帧只要在 SYNC0 之前到就行。真实运行里驱动器的周期离散约 5 纳秒。前提是帧真的提前到了；晚于 SYNC0 会怎样，第 5 集再讲。

The joints no longer act when a frame arrives; they act on SYNC0 from their own clocks. The frame only has to arrive before SYNC0. In the real run the drives' period spread is about 5 ns. That assumes the frame does arrive first; what happens if it is late is Episode 5.

依据 · Evidence

同一条虚拟总线、同一台 Mac，只把驱动器改成按 SYNC0 动作；三次运行周期离散都约 5 纳秒（模拟芯片）。 / Same virtual line, same Mac, only the drives switched to act on SYNC0; all three runs show about 5 ns period spread (simulated chips).

`episodes/ep01-sync-is-accuracy/data/runs.sh`